

MediAssist AI

Project Report: AI Healthcare Assistant using Hugging Face & Prompt Engineering

Submitted by: Shivansh Kumar

Roll No.: 2301921520178

Institute: G L Bajaj Institute of Technology and Management

1. Introduction

MediAssist AI Pvt. Ltd.'s existing chatbot suffered from four core problems: inconsistent responses across similar queries, hallucinated medical information, unstructured and hard-to-parse outputs, and rigid, poorly maintained prompts that could not adapt across models or user roles. This project addresses each of these problems by rebuilding the assistant on Hugging Face Transformers and applying deliberate prompt engineering techniques, rather than relying on default, unstructured queries to the model.

2. Model Selection

Primary model: Microsoft/Phi-3-mini-4k-instruct (3.8B parameters). Chosen as the main chatbot engine because it is instruction-tuned out of the box, small enough to run on Colab's free-tier T4 GPU, and produces coherent, well-structured long-form responses important for tasks like discharge summaries and multi-section disease overviews.

Secondary model: Qwen/Qwen2.5-1.5B-Instruct (1.5B parameters). Loaded for the model comparison task, roughly a third the size of Phi-3-mini, used to evaluate the trade-off between speed/resource usage and response depth.

Both models were accessed through the standard Auto Tokenizer / AutoModelForCausalLM / pipeline interface, so switching models required changing only a single variable satisfying the requirement that the chatbot support switching between different LLMs without rewriting application logic.

3. Prompt Engineering Techniques Used

Technique	Where applied	Purpose
Role Prompting	Task 2, 4, 7, Bonus	Frames the model as a specific persona so it stays within an appropriate scope of authority
Context Injection	Task 2, 4, 7	Grounds responses in specific case details instead of generic priors
Constraints	Task 2, 4, 5, 7	Enforces word limits, forbids definitive diagnosis claims, restricts to evidence-based info
Output Formatting	Task 2, 4, 5, 6, Bonus	Forces Markdown or JSON structure for consistent, parseable output
Few-Shot Prompting	Task 3	Worked examples teach the exact output pattern before the real input
Chain-of-Thought (hidden)	Task 4	Internal step-by-step reasoning without exposing it in the final output
Role-Adaptive Templating	Bonus	One model, swappable instruction block per user role

4. Challenges Faced

- **Hallucination under vague prompts:** the bare prompt "Tell me about diabetes" produced generic, inconsistent output; adding role/context/constraints tightened it to the requested structure.
- **Unreliable JSON output:** the model sometimes added text around the JSON block despite instructions, requiring defensive regex extraction before parsing.
- **Reasoning leakage:** early Task 4 attempts exposed full step-by-step reasoning in the output; explicitly separating "internal reasoning" from "output" fixed this.
- **Model size vs. quality trade-off:** Qwen2.5-1.5B was faster and lighter but less nuanced on multi-step reasoning prompts than Phi-3-mini.

5. Improvements Made After Prompt Refinement

- Role + context + constraint + format structuring (Task 2) produced consistently sectioned, on-topic responses.
- Three worked examples (Task 3) eliminated ambiguity in the expected summarization format.

- Separating hidden reasoning from output (Task 4) stopped raw chain-of-thought from leaking into doctor-facing summaries.
- Constraining Task 7's prompt to a ranked, test-backed differential turned a low-confidence guess-list into an actionable, safety-framed response.
- The role-based template (Bonus) confirmed one model can serve multiple hospital user types by swapping only the instruction block.

6. Conclusion

Restructuring the assistant around explicit prompt engineering techniques rather than passing raw queries directly to the model addressed all four original problems: responses became more consistent, hallucinations were reduced through grounding and constraints, outputs were reliably structured, and the pipeline design allowed switching between Hugging Face models with minimal code changes. This confirms that in a regulated domain like healthcare, prompt design is as critical to output quality and safety as model selection itself.